Register / Shift Register

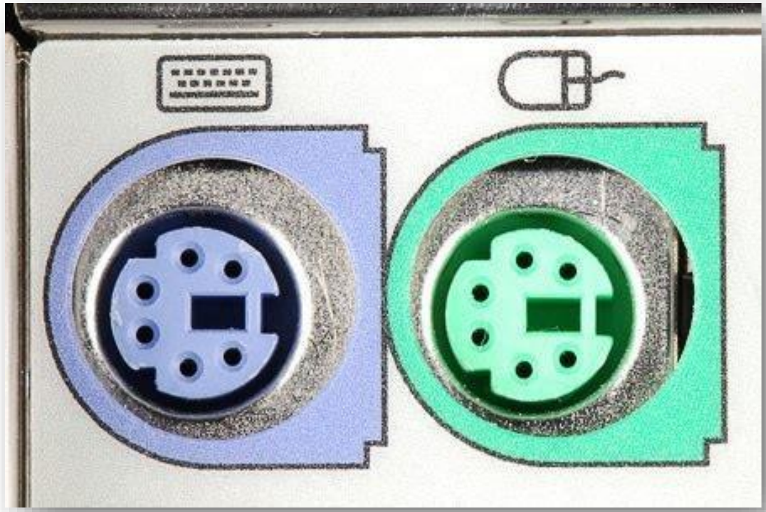
Logic Design of Digital Systems (300-1209) section 1

LECTURE 11

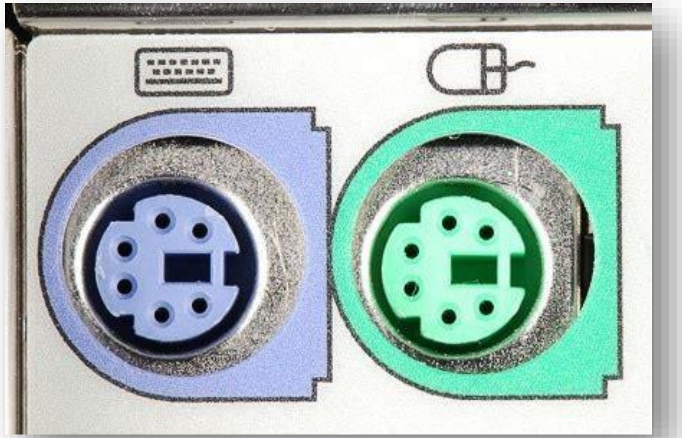
KRISADA PHROMSUTHIRAK

Outline

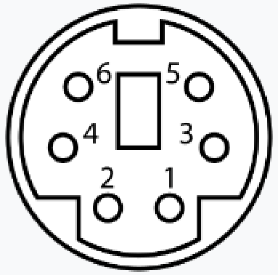
- ❑ Serial In – Serial Out (SISO) shift register
- ❑ Serial In – Parallel Out (SIPO) shift register
- ❑ Parallel In – Serial Out (PISO) shift register
- ❑ Parallel In – Parallel Out (PIPO) shift register



- ❑ Serial Adder (วงจรบวกแบบอนุกรม)



Pinout

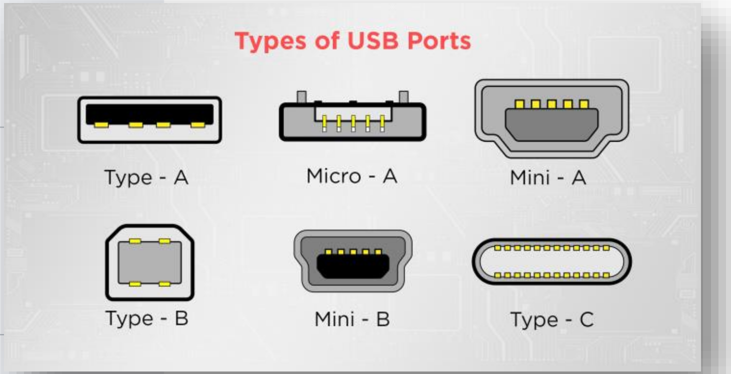


Female connector from the front

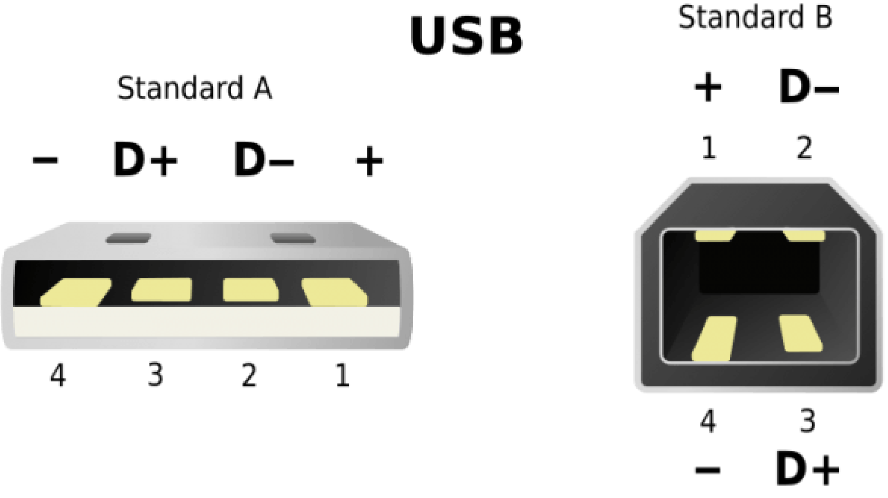
Pin 1	+DATA	Data
Pin 2		Not connected[b]
Pin 3	GND	Ground
Pin 4	Vcc	+5 V DC at 275 mA
Pin 5	+CLK	Clock
Pin 6		Not connected[c]

Available connectors by USB standard

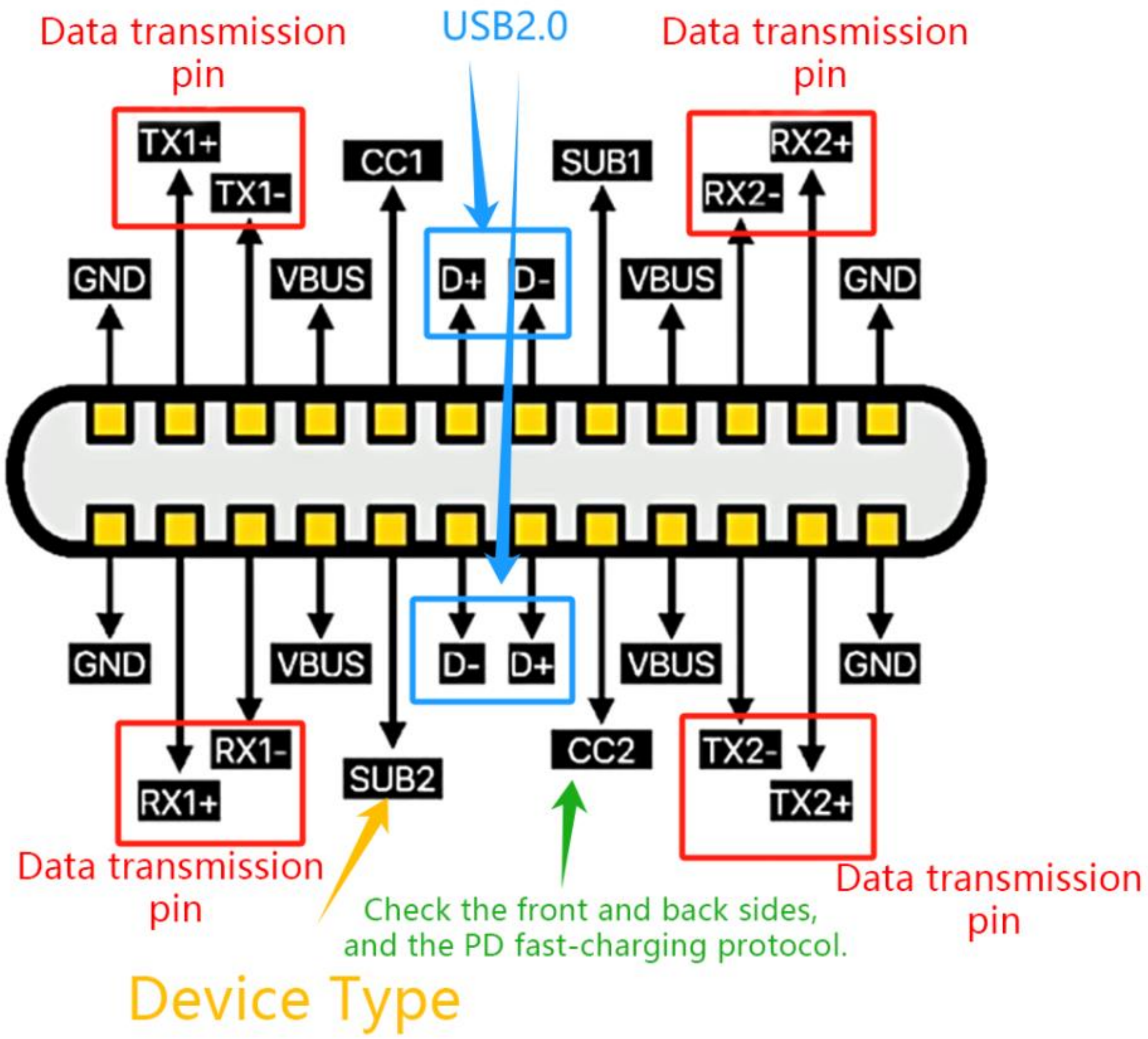
Standard		USB 1.0 1996	USB 1.1 1998	USB 2.0 2000	USB 2.0 Revised	USB 3.0 2008	USB 3.1 2013	USB 3.2 2017	USB4 2019	USB4 2.0 2022
Max Speed	Recommended marketing names from 2022[16]	Basic-Speed		High-Speed		USB 5Gbps	USB 10Gbps	USB 20Gbps	USB 40Gbps	USB 80Gbps
	Original label	*Low-Speed & Full-Speed*				*SuperSpeed, or SS*	*SuperSpeed+, or SS+*	*SuperSpeed USB 20Gbps*		
	Operation mode					USB 3.2 Gen 1×1	USB 3.2 Gen 2×1	USB 3.2 Gen 2×2	USB4 Gen 3×2	USB4 Gen 4×2
	Signaling rate	1.5 Mbit/s & 12 Mbit/s		480 Mbit/s		5 Gbit/s	10 Gbit/s	20 Gbit/s	40 Gbit/s	80 Gbit/s
Connector	Standard-A	Type-A 1.0 - 1.1		Type-A 2.0		Type-A SuperSpeed		Type-A SuperSpeed [rem 1]		
	Standard-B	Type-B				Type-B SuperSpeed		Type-B SuperSpeed [rem 1]		
	Mini-A	[rem 2]		Mini-A		—				
	Mini-AB[rem 3][rem 4]			Mini-AB						
	Mini-B			Mini-B						
	Micro-A[rem 5]	[rem 2][rem 6]			Micro-A	Micro-A SuperSpeed		Micro-A SuperSpeed [rem 1]	—	
	Micro-AB[rem 3][rem 7]				Micro-AB	Micro-AB SuperSpeed		Micro-AB SuperSpeed [rem 1]		
	Micro-B				Micro-B	Micro-B SuperSpeed		Micro-B SuperSpeed [rem 1]		
	Type-C (USB-C)	[rem 6]			(Enlarged to show detail)					



USB



Standard A
– D+ D– +
4 3 2 1
Standard B
+ D–
1 2
4 3
– D+



USB2.0
Data transmission pin
TX1+
TX1-
CC1
SUB1
RX2+
RX2-
GND
VBUS
D+
D-
VBUS
GND
GND
VBUS
D-
D+
VBUS
GND
RX1-
RX1+
SUB2
CC2
TX2-
TX2+
Data transmission pin
Data transmission pin
Data transmission pin
Check the front and back sides, and the PD fast-charging protocol.
Device Type

Register

Flip-flop is a 1-bit memory cell which can be used for storing the digital data. To increase the storage capacity in terms of number of bits, we have to use a group of flip-flop. Such a group of flip-flop is known as a Register. The n-bit register will consist of n number of flip-flop, and it can store an **n-bit** word.

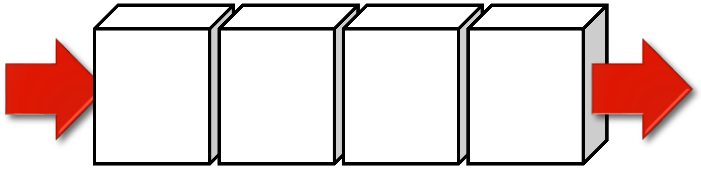
The information stored within these registers can be transferred with the help of **shift registers**. Shift Register is a group of flip flops used to store multiple bits of data. The bits stored in such registers can be made to move within the registers and in/out of the registers by applying clock pulses.

Shift Register

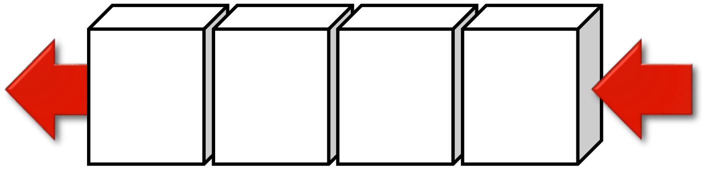
An n-bit shift register can be formed by connecting n flip-flops where each flip flop stores a single bit of data.

- The registers which will shift the bits to left are called “Shift left registers”
- The registers which will shift the bits to right are called “Shift right registers”

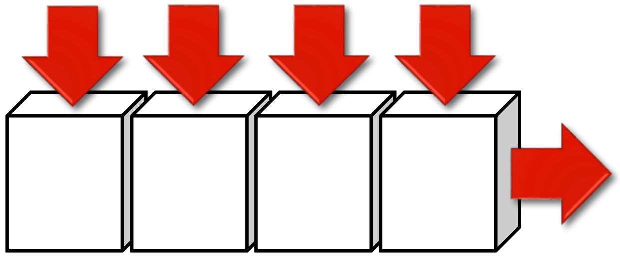
Shift Register



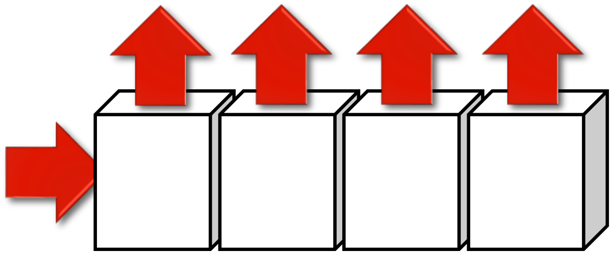
serial in | shift-right | serial out



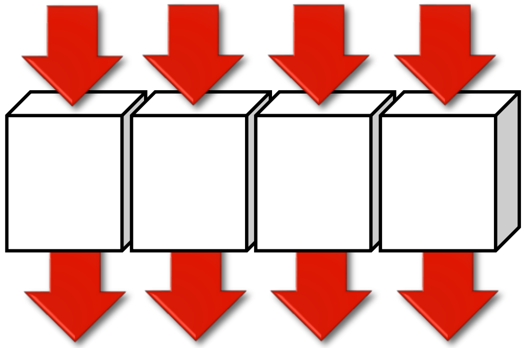
serial in | shift-left | serial out



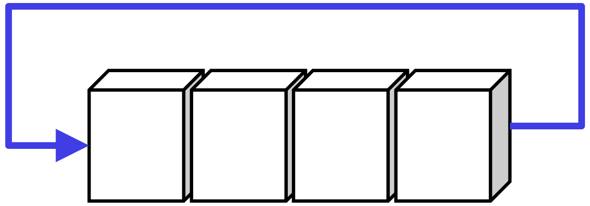
parallel in | serial out



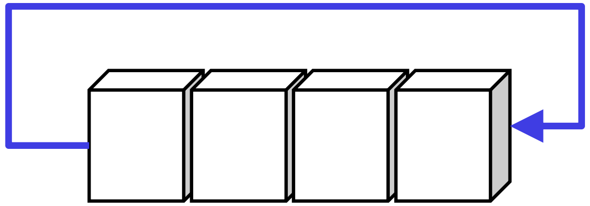
serial in | parallel out



parallel in | parallel out

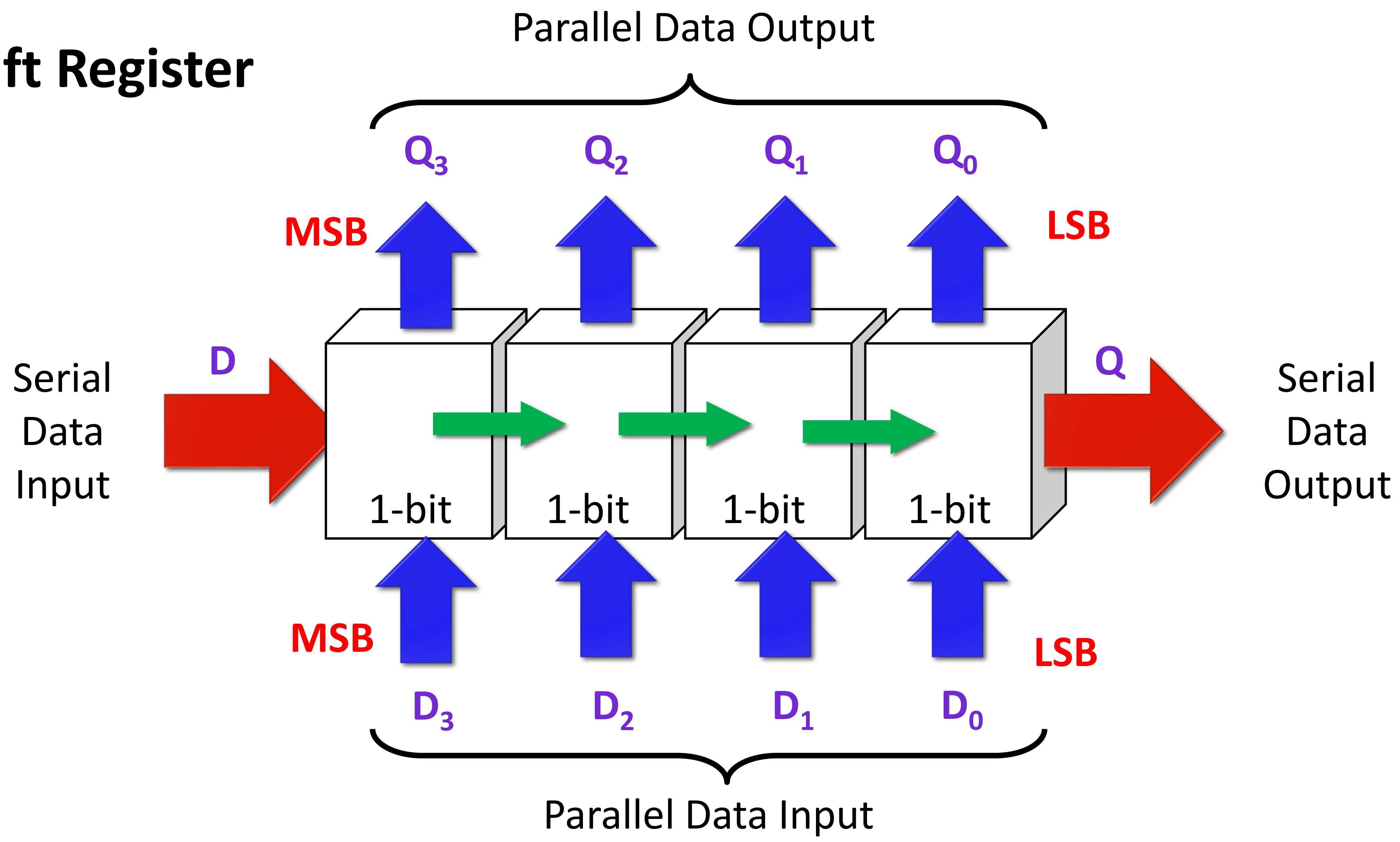


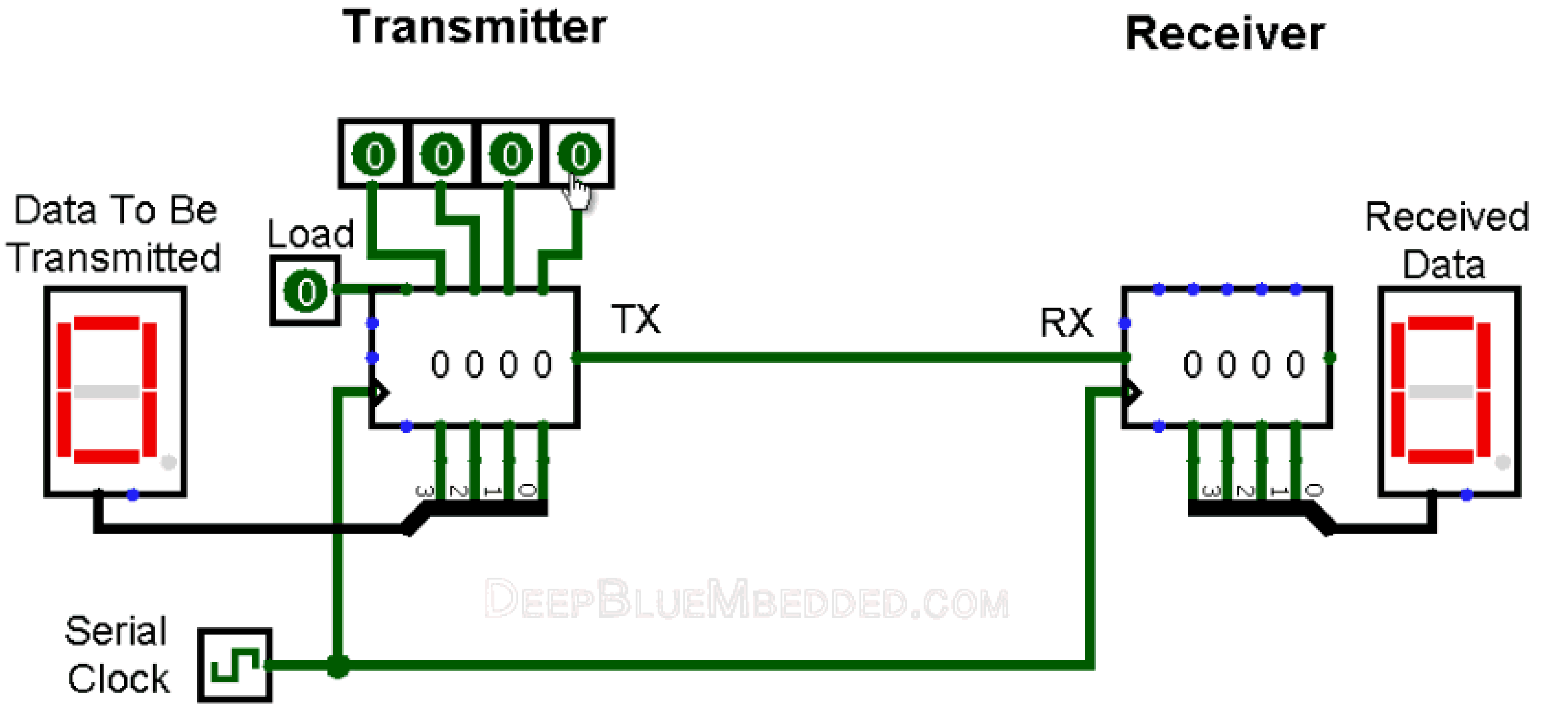
Rotate right



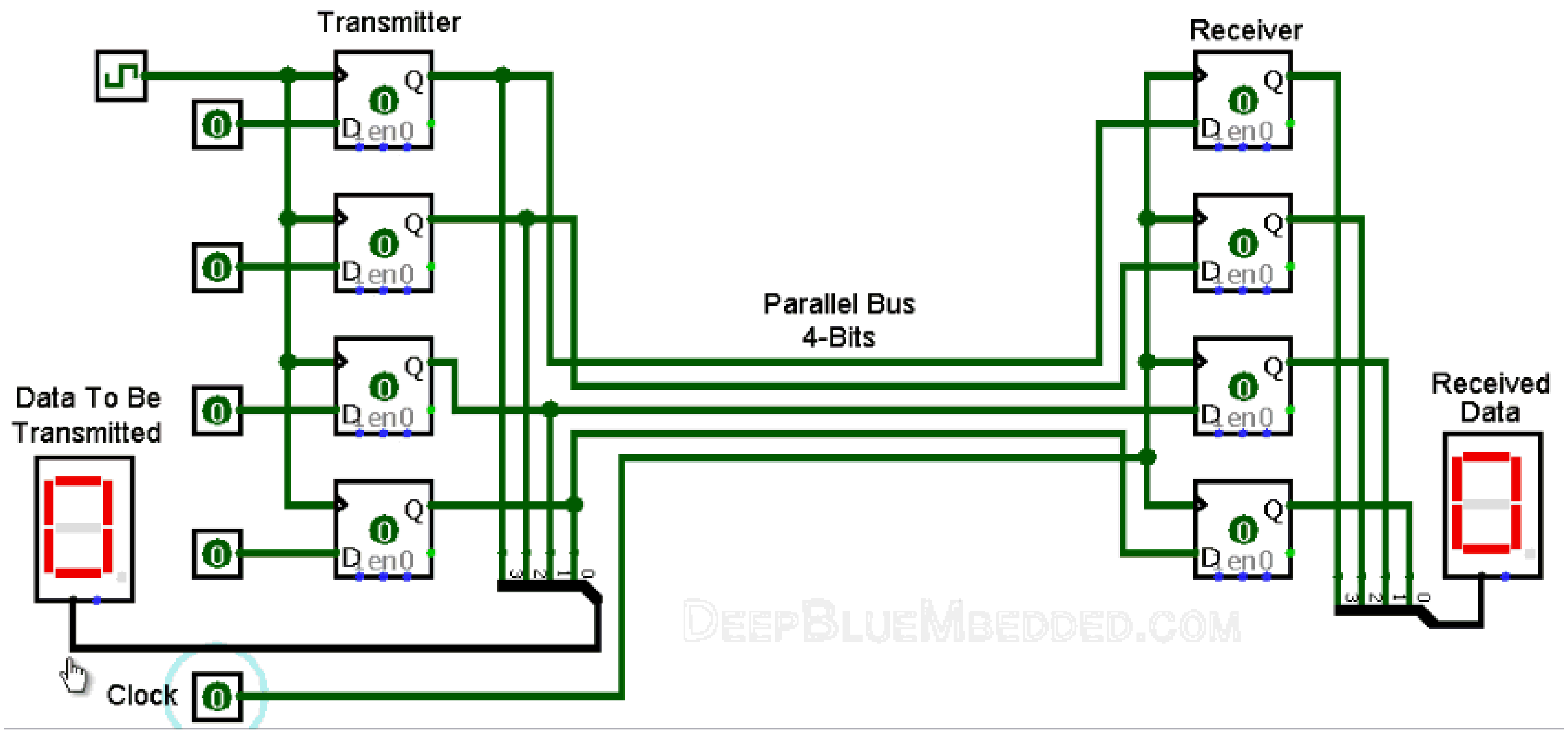
Rotate left

Shift Register





REF: https://deepbluembedded.com



REF: https://deepbluembedded.com

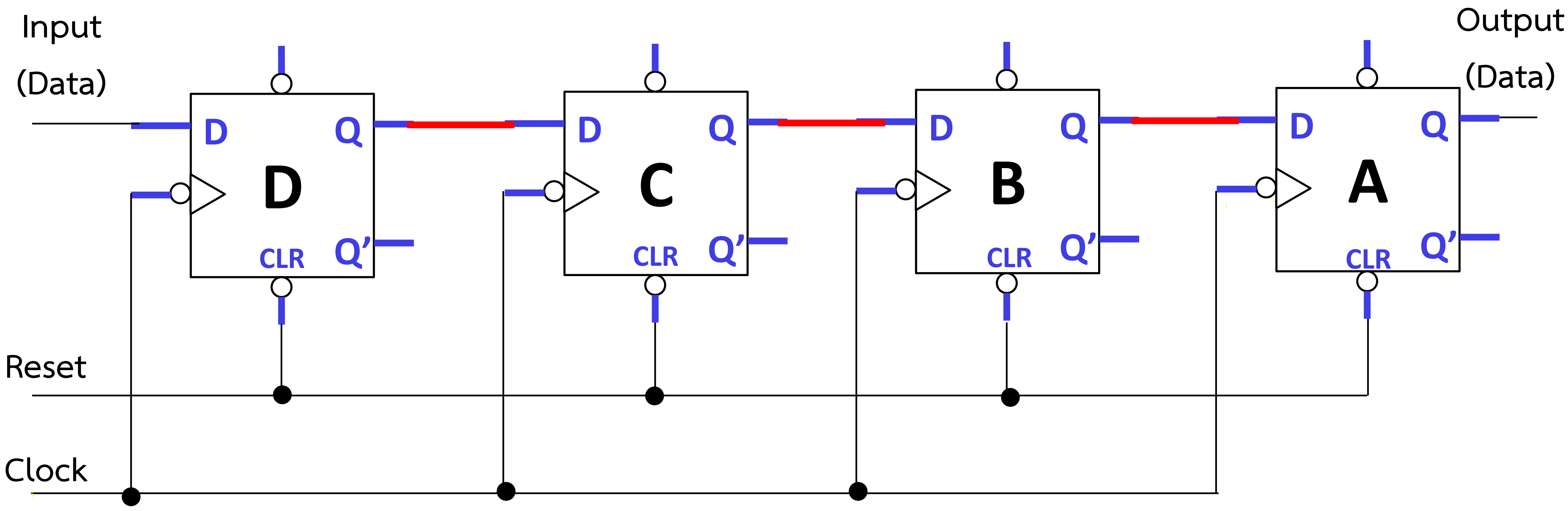
Feature	Serial Communication	Parallel Communication
Speed	Slow	Fast
Distance Range	Much Farther	Shorter
Transfer Method	One bit at a time	One Bytes or more at a time
Wires Inside	Few wires	Multiple, more wires, each bit has a dedicated wire.
Applications	Computer small modules interfacing. Simple data. (such as Mouse)	Short distance High-Speed. (such as Printer)

Feature	Serial Communication	Parallel Communication
Cables Pictures	  	  

[1] Serial In – Serial Out (SISO)

The shift register, which allows serial input (one bit after the other through a single data line) and produces a serial output is known as Serial-In Serial-Out shift register. Since there is only one output, the data leaves the shift register one bit at a time in a serial pattern, thus the name Serial-In Serial-Out Shift Register.

Example: 4-bits SISO shift register

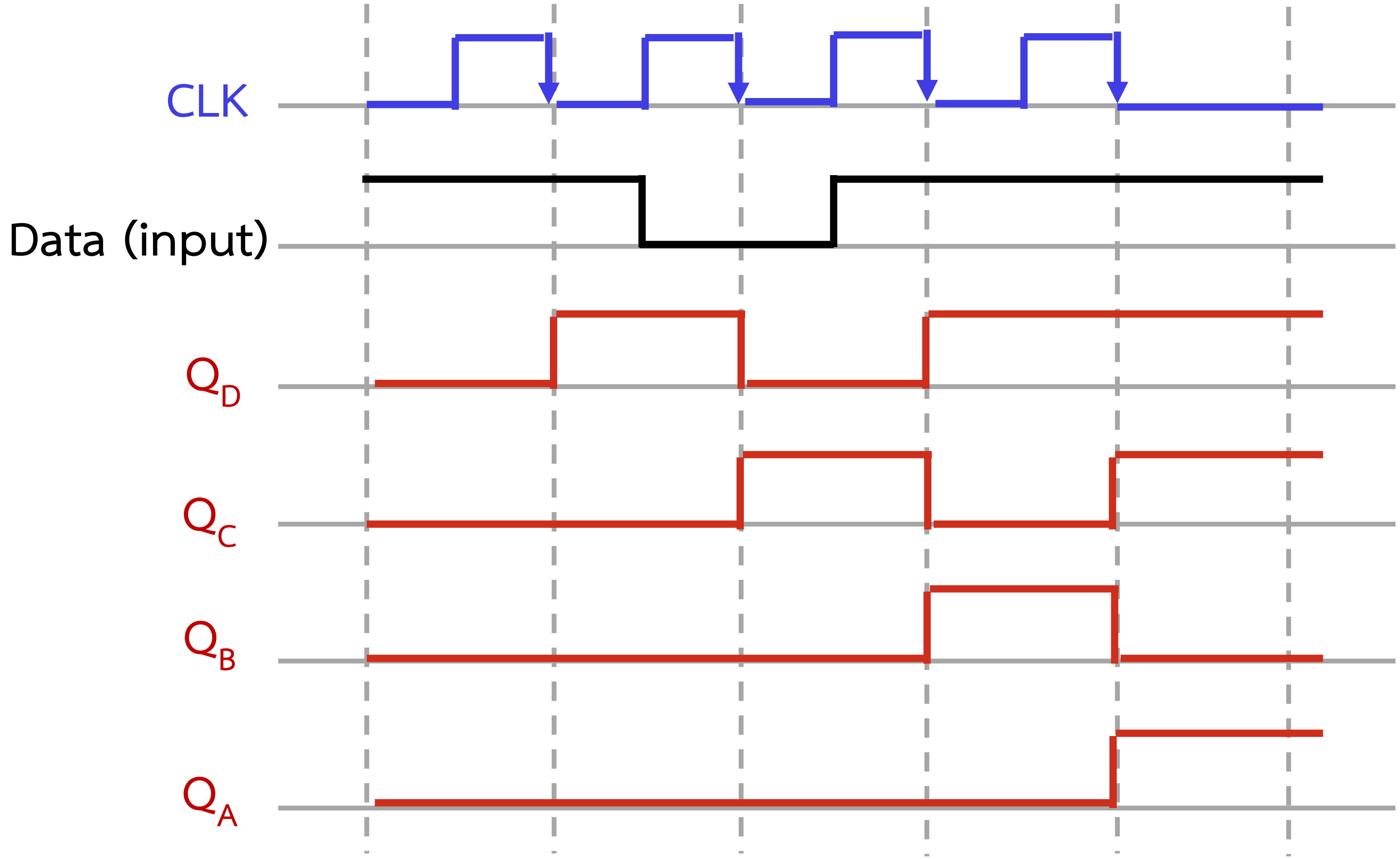


Example: 4-bits SISO shift register

Clock	Input	Data in Register			
		Q_D	Q_C	Q_B	Q_A
0	0	0	0	0	0
1	1	1	0	0	0
2	0	0	1	0	0
3	1	1	0	1	0
4	1	1	1	0	1

Operation Table

Example: 4-bits SISO shift register

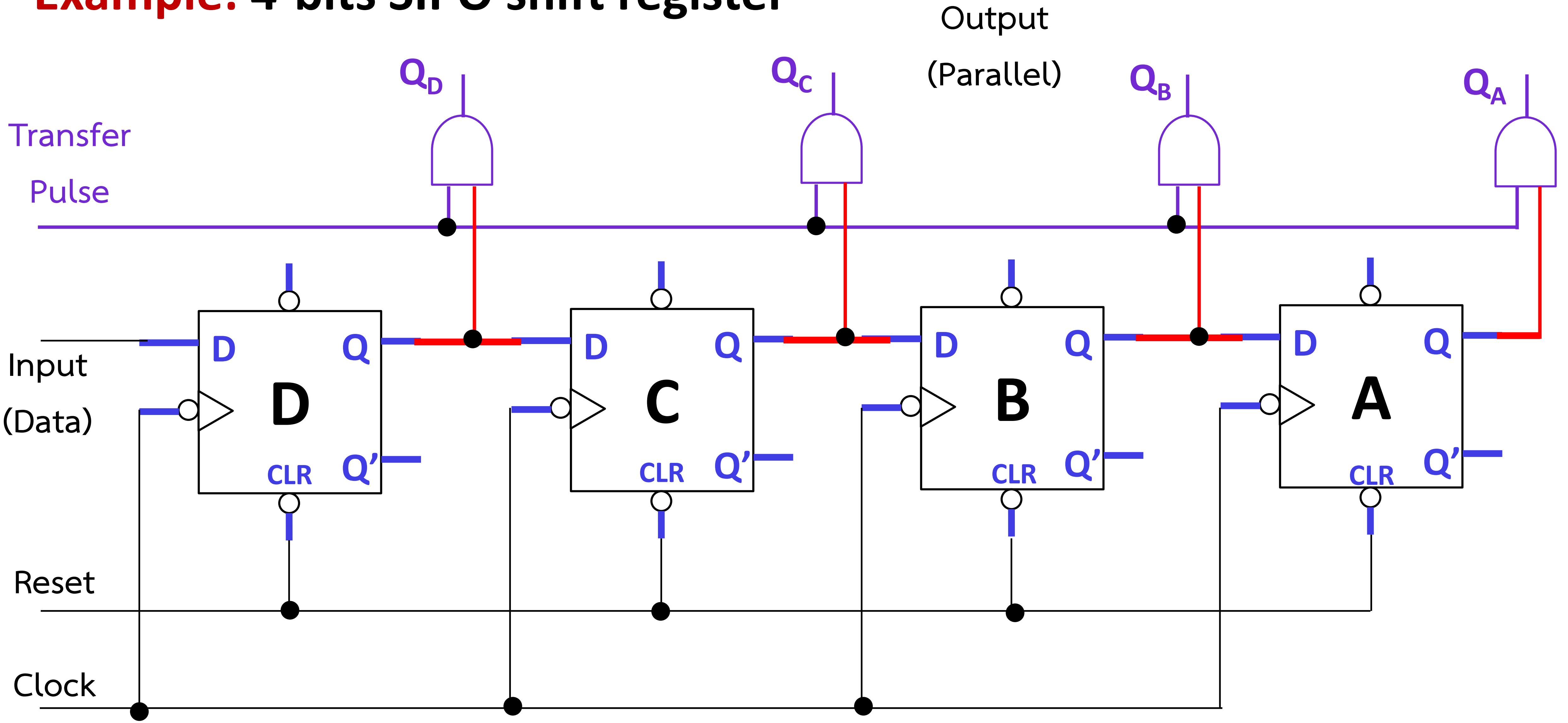


Timing Diagram

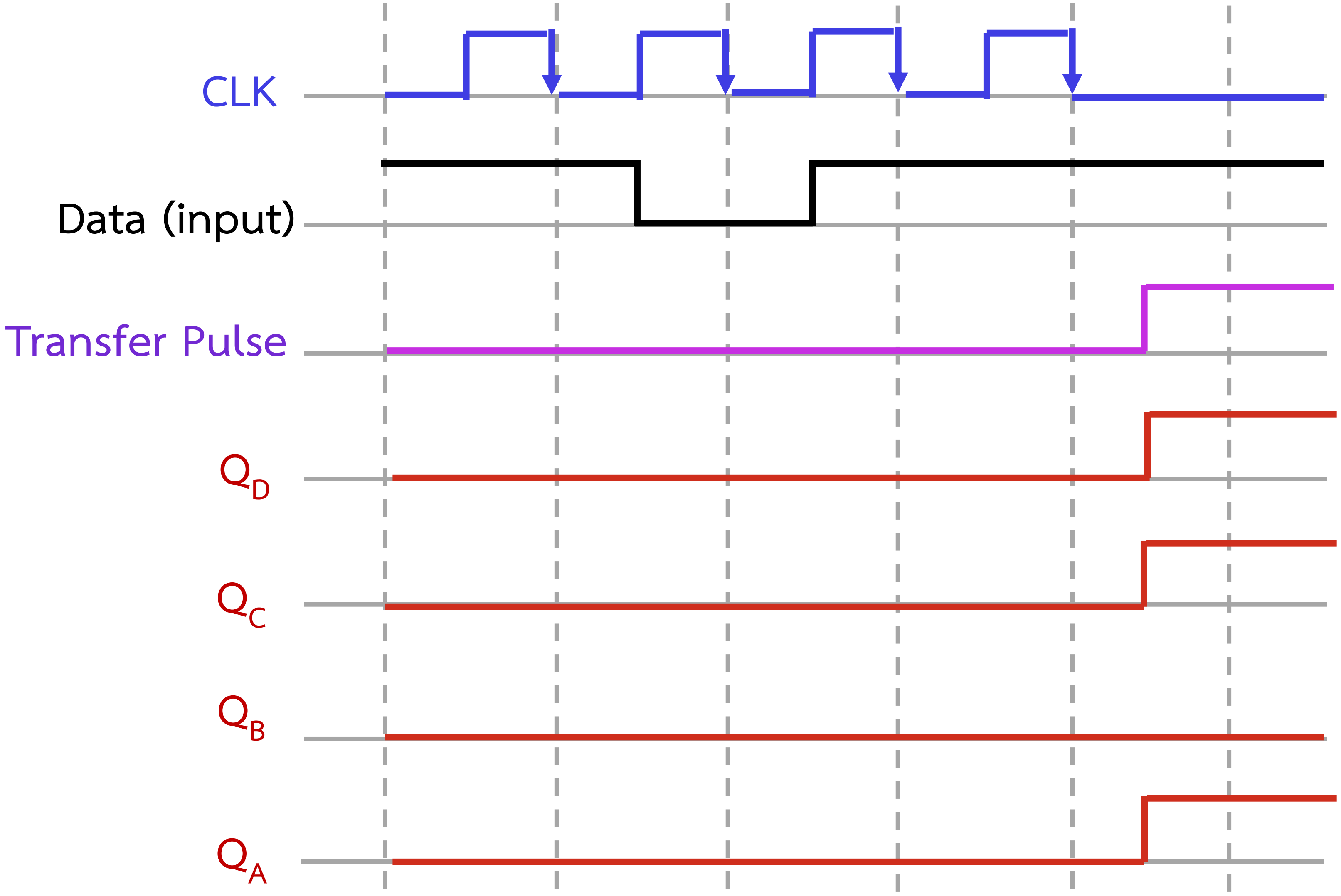
[2] Serial In – Parallel Out (SIPO)

The shift register, which allows serial input (one bit after the other through a single data line) and produces a parallel output is known as Serial-In Parallel-Out shift register.

Example: 4-bits SIPO shift register



Example: 4-bits SIPO shift register

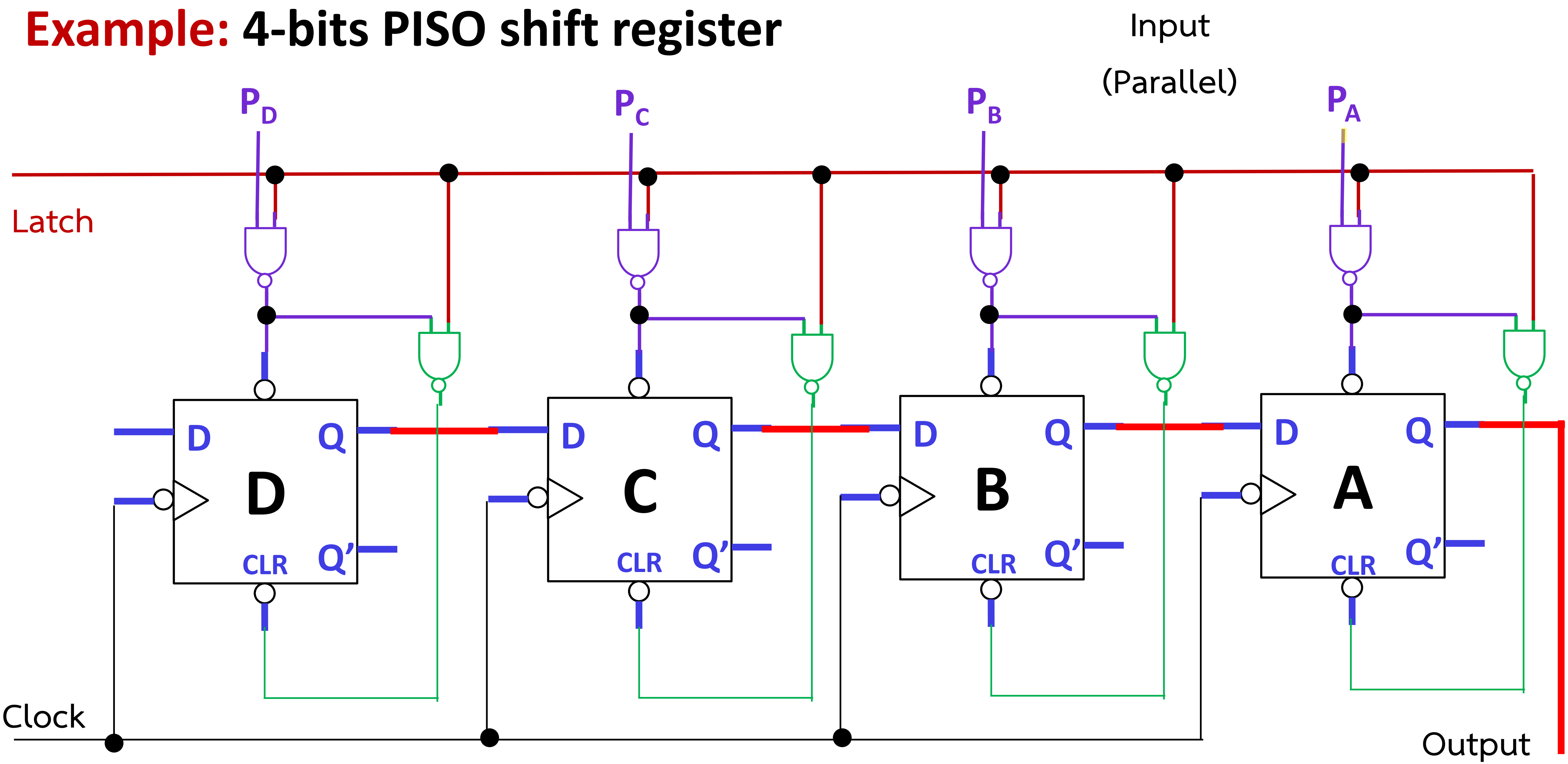


Timing Diagram

[3] Parallel In – Serial Out (PISO)

The shift register, which allows parallel input (data is given separately to each flip flop and in a simultaneous manner) and produces a serial output is known as Parallel-In Serial-Out shift register.

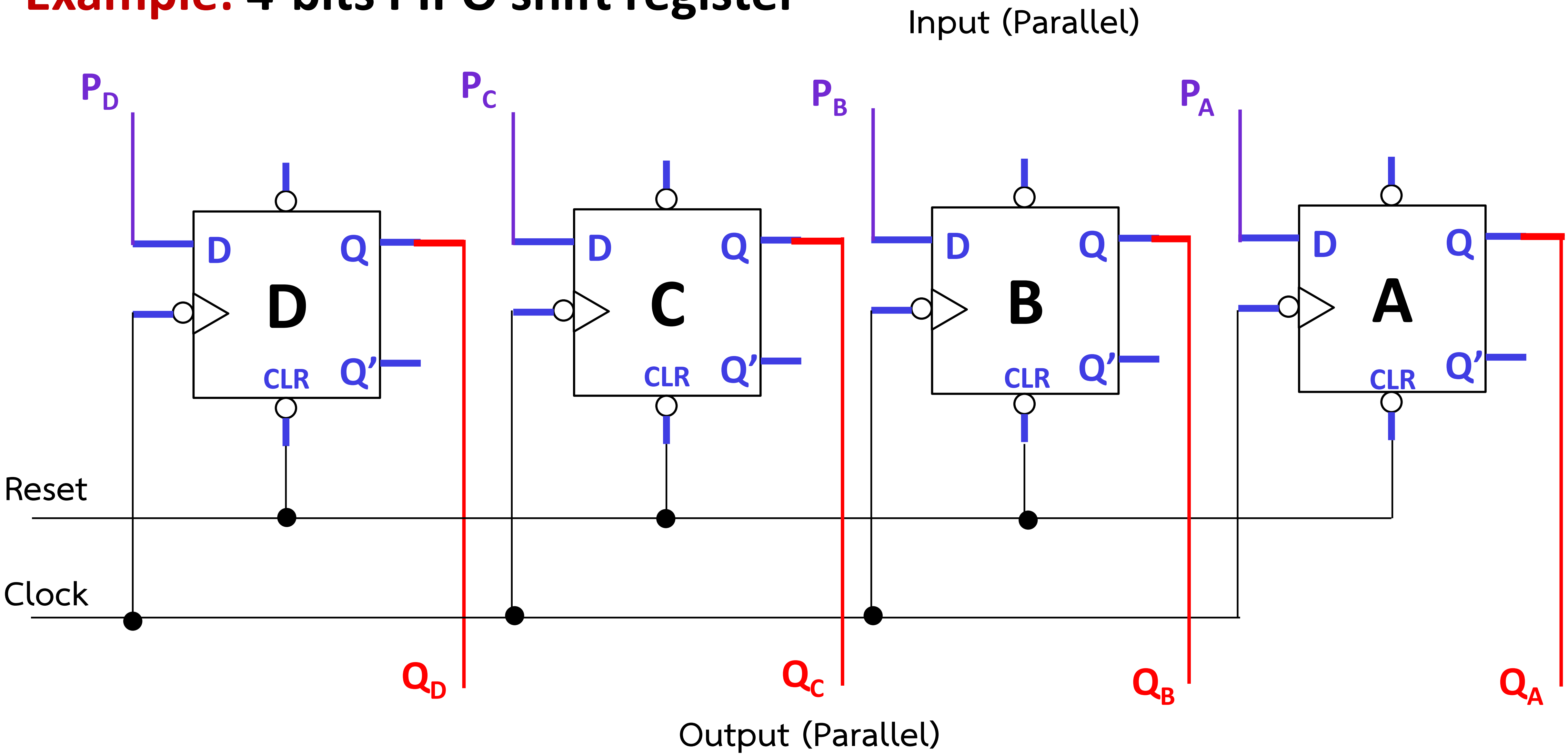
Example: 4-bits PISO shift register



[4] Parallel In – Parallel Out (PIPO)

The shift register, which allows parallel input (data is given separately to each flip flop and in a simultaneous manner) and also produces a parallel output is known as Parallel-In parallel-Out shift register.

Example: 4-bits PIPO shift register



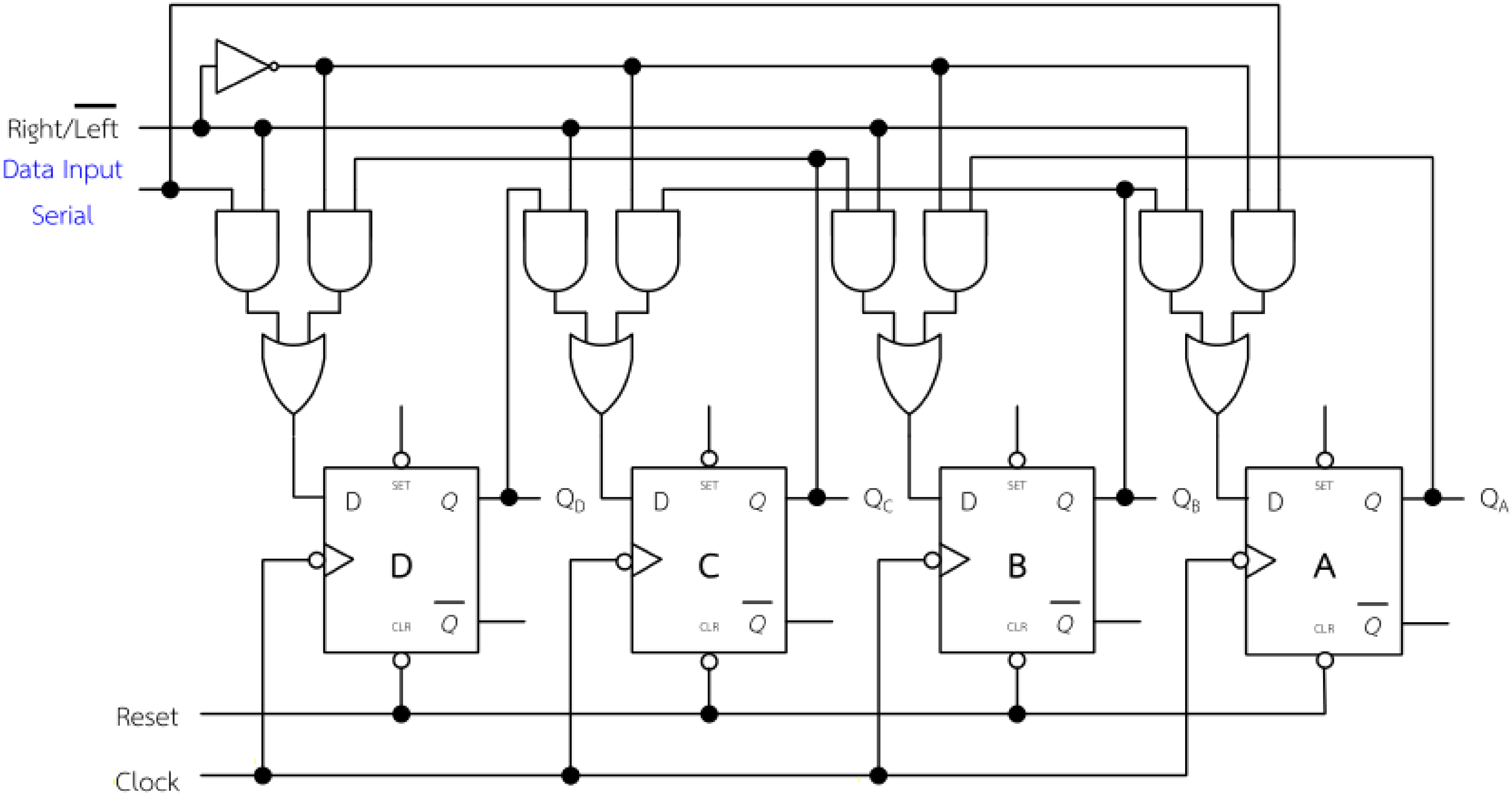
Take a Break

RELAX TIME

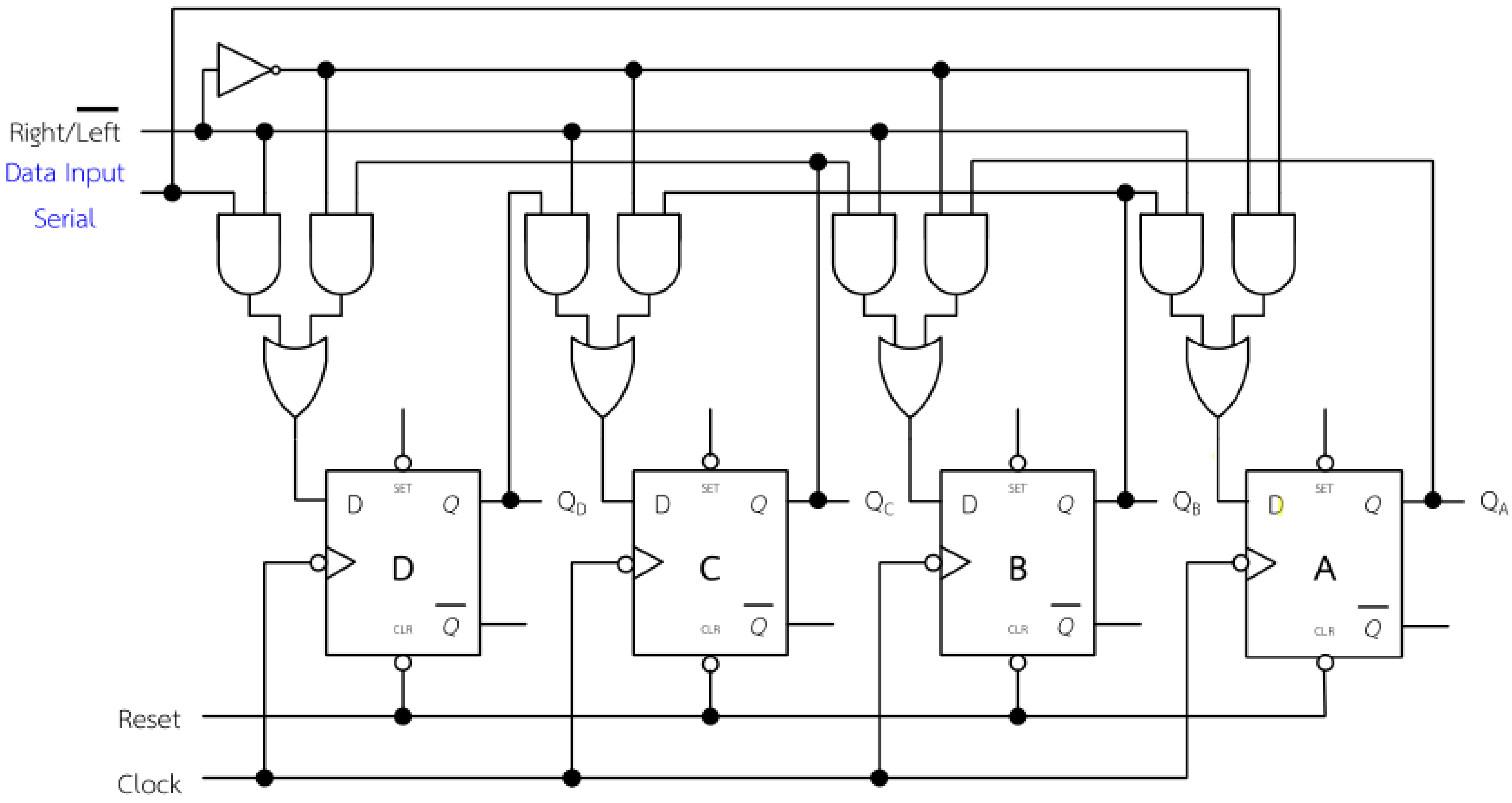
Bidirectional Shift Register

Bidirectional shift registers are the registers which are capable of shifting the data either right or left depending on the mode selected. If the mode selected is 1(high), the data will be shifted towards the right direction and if the mode selected is 0(low), the data will be shifted towards the left direction.

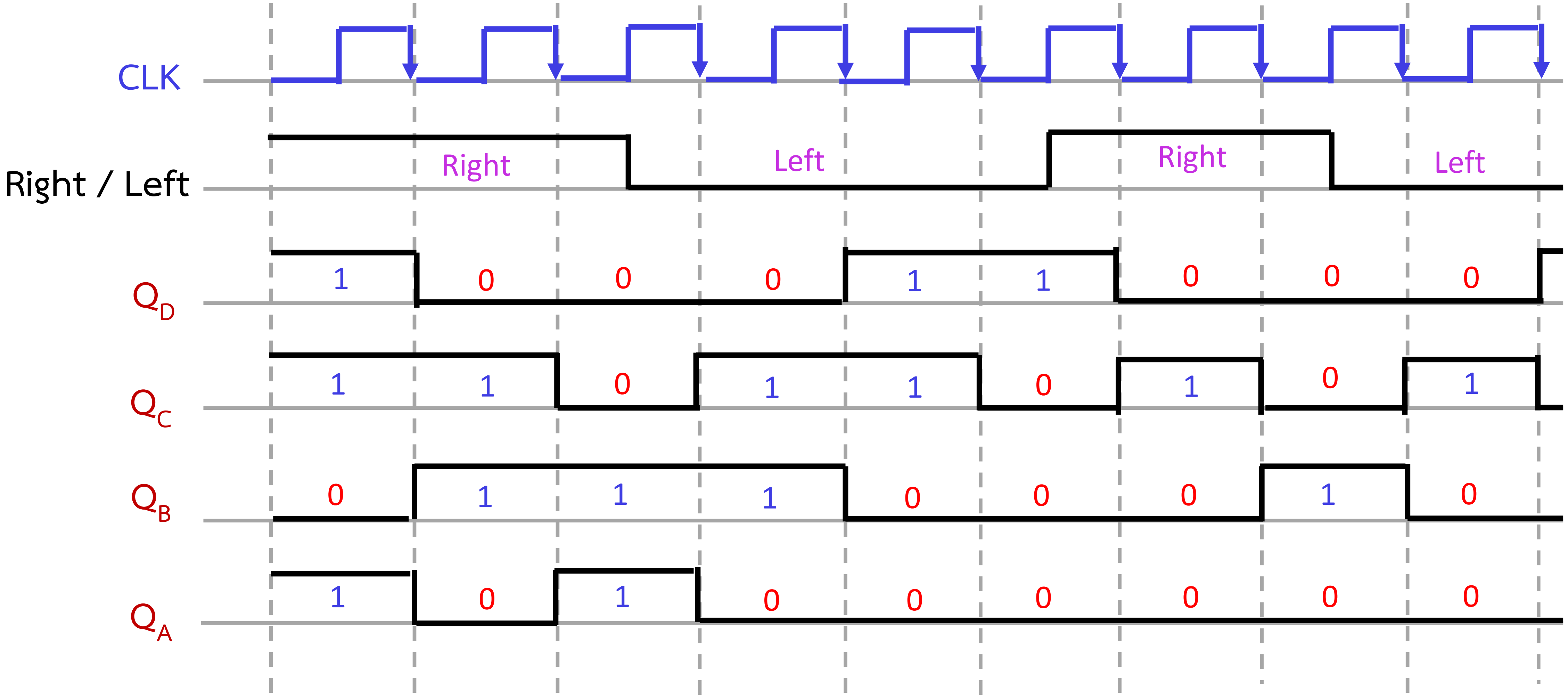
Example: 4-bits Bidirectional Shift Register



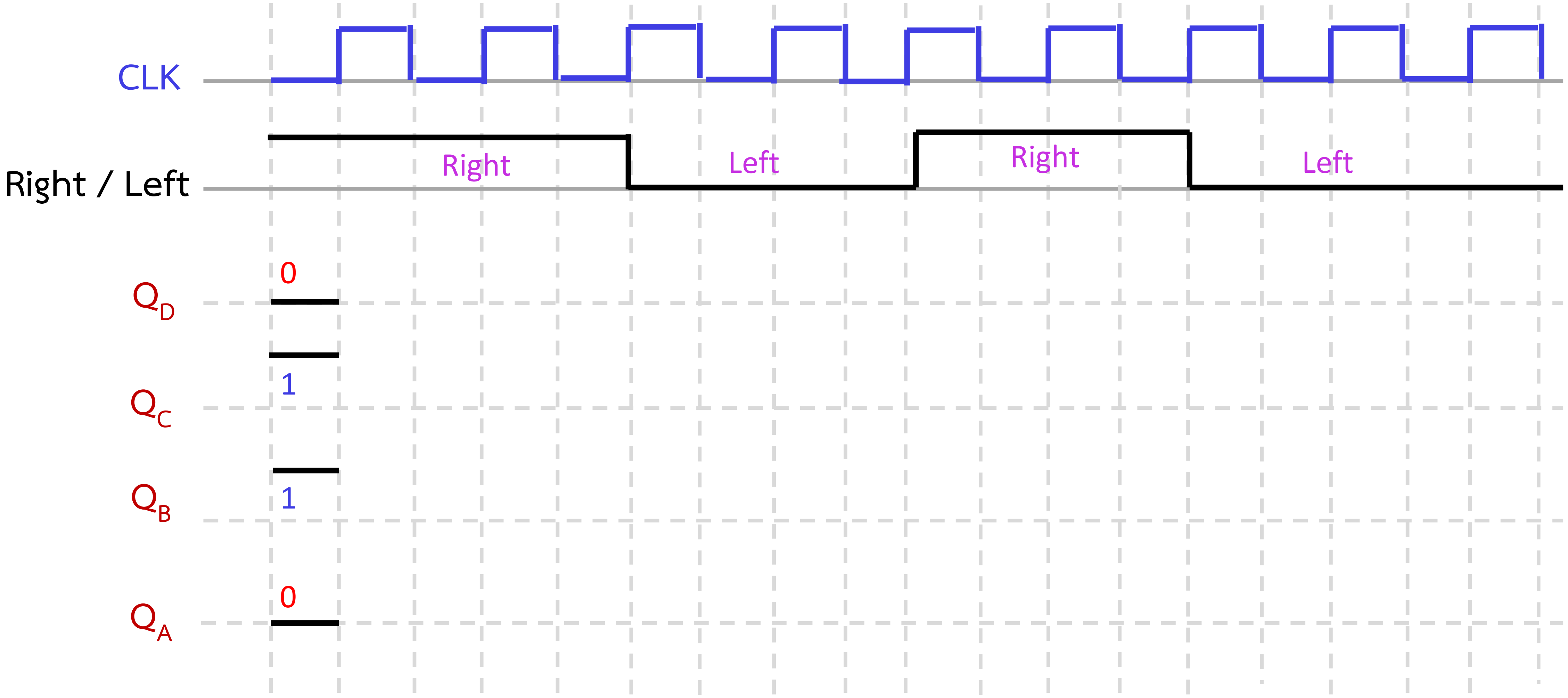
Example: 4-bits Bidirectional Shift Register



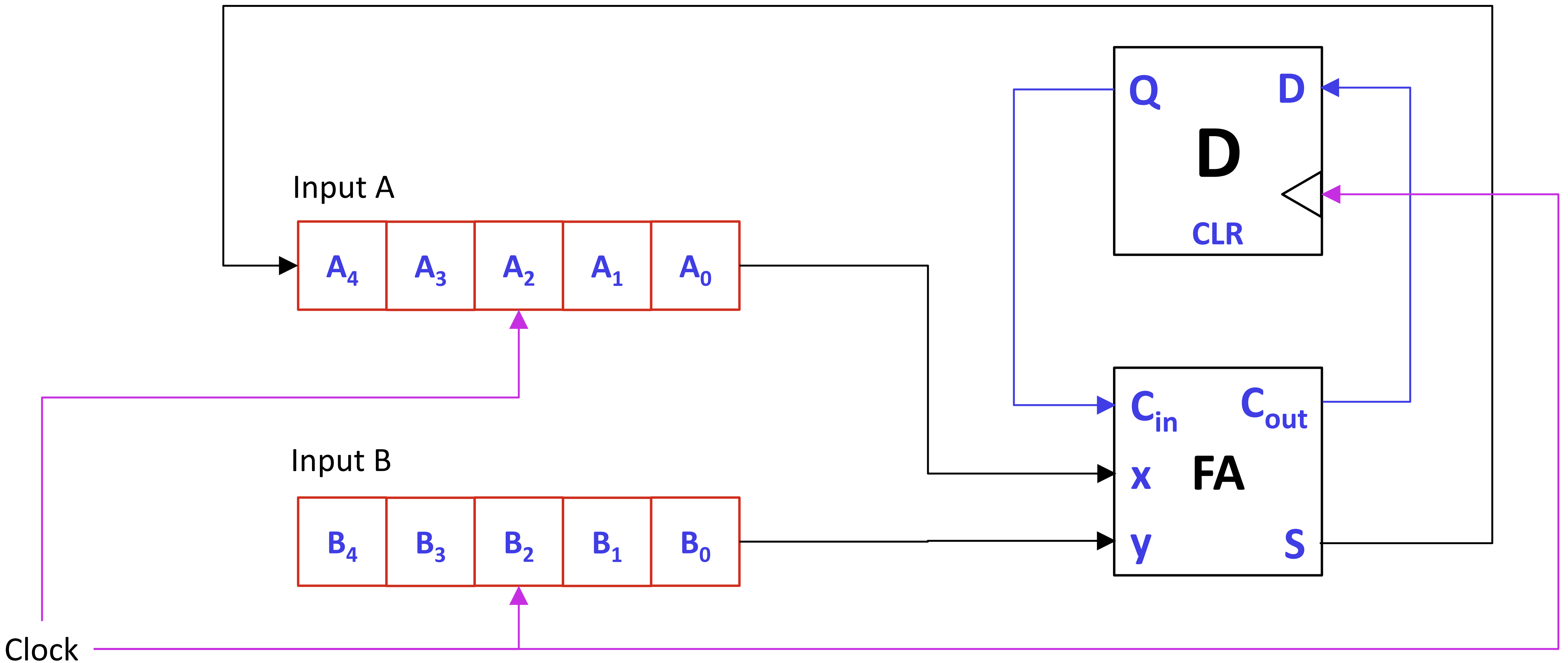
Example: 4-bits Bidirectional Shift Register



Quiz: Please complete the timing diagram of 4-bits Bidirectional Shift Register

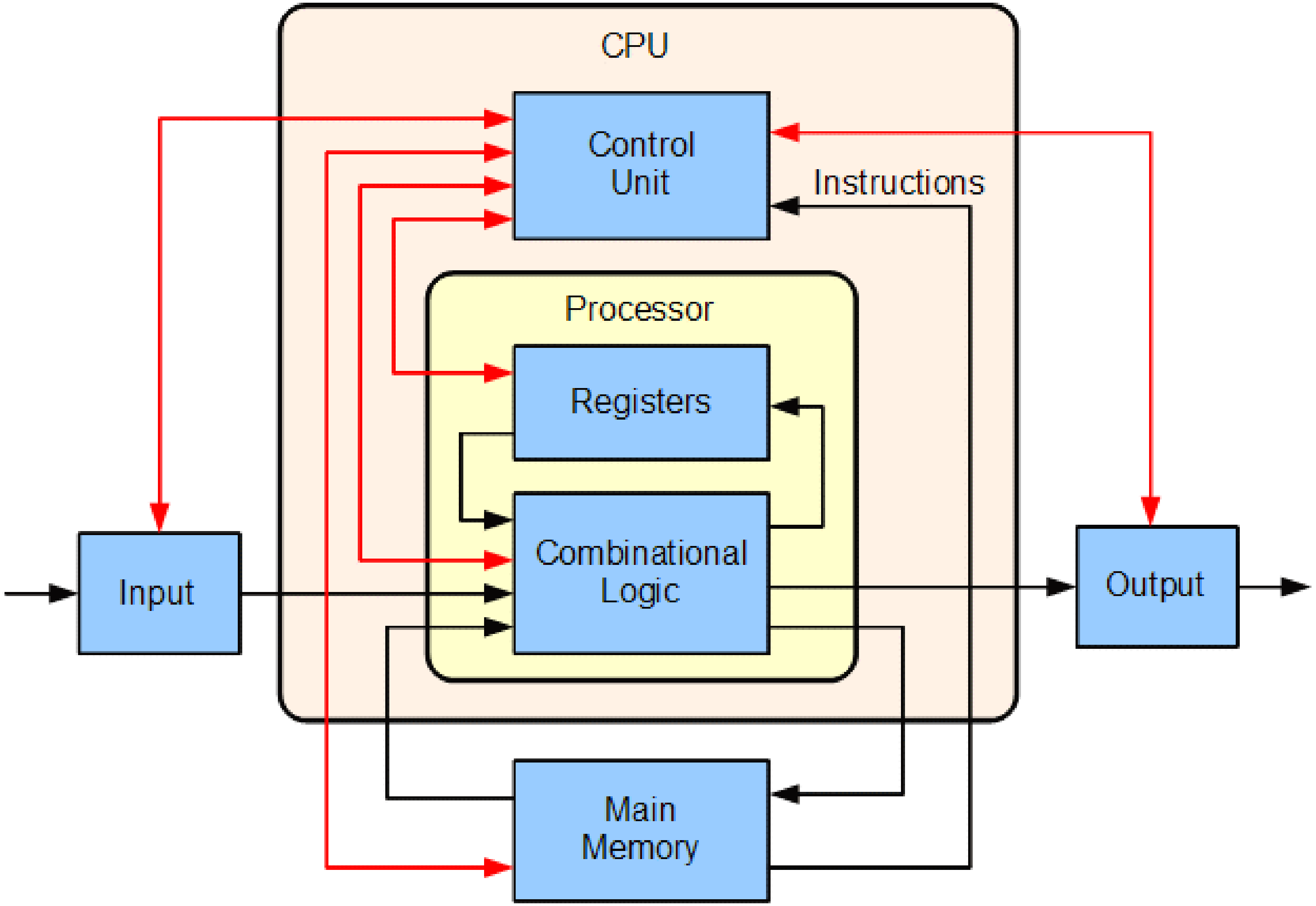


Serial Adder



REF: https://www.geeksforgeeks.org/serial-binary-adder-in-digital-logic/

END



Basic computer with uniprocessor CPU